

Halo Mass Mediates the Colour–Environment Relation in GAMA: Evidence from Partial Correlation and Distribution Shape Analysis

K. Sloan

Independent Researcher — working draft for MNRAS Letters

Abstract

We investigate the relationship between galaxy colour ($u-r$) and local environment for galaxies in the GAMA DR4 survey, using partial correlation to control for group halo mass and bootstrap null tests to assess physical evolution in the shape of the colour distribution with redshift. For elliptical galaxies ($N = 1,270$), raw correlations between $u-r$ and all four environment metrics collapse to $r \approx 0$ once group halo mass is controlled, demonstrating complete mediation. For the full galaxy population ($N = 22,531$), $\sim 50\%$ of the signal survives halo control, indicating an independent morphology-linked pathway. Skewness and Box-Cox λ of the $u-r$ distribution evolve monotonically with redshift (Spearman $|r| = 1.00$, $p < 0.001$ for both): skewness shifts from $+0.90$ at $z = 0.03$ to -0.82 at $z = 0.33$, and λ from -0.20 to $+2.70$. Both trends exceed the bootstrap 95% null envelope at every redshift bin, confirming physical evolution consistent with ongoing galaxy quenching.

1. Introduction

The correlation between galaxy colour and local environment is among the most robust observational results in galaxy evolution (Dressler 1980; Blanton & Moustakas 2009). The outstanding question is mechanistic: does environment directly drive quenching, or is the signal primarily a reflection of halo mass, which correlates with both environment and the efficiency of quenching processes such as ram-pressure stripping and strangulation?

A complementary question is whether the shape of the colour distribution encodes evolutionary information beyond the mean. If the $u-r$ distribution transforms systematically with redshift in ways exceeding sampling noise, that transformation constrains the quenching timescale independently of correlation analyses.

We address both questions using GAMA DR4 (Driver et al. 2022), exploiting its group catalogue (Robotham et al. 2011) and environment measures (Brough et al. 2013).

2. Data and Sample

GAMA DR4 StellarMassesLambdarv24 (200,077 galaxies), quality cuts $0.5 < u-r < 4.0$, $\log M_* > 8.0$, $0.002 < z < 0.35$: 165,488 galaxies for shape analysis. Partial correlation additionally uses G3CGalv10 (group membership), G3CFoFGroupv10 (halo mass proxy MassA), and EnvironmentMeasuresv06 (CountInCyl, DistanceTo5nn, SurfaceDensity,

AGEDenPar). Elliptical subsample: VisualMorphologyv03 ELLIPTICAL_CODE = 1, N = 1,270. Full group population N = 22,531.

3. Methods

3.1 Partial Correlation

We residualise both $u-r$ and each log-environment measure against $\log(\text{MassA})$ via linear regression and correlate residuals (Pearson and Spearman). Results at individual galaxy level and group level (mean $u-r$; red fraction $f_{\text{red}} = \text{fraction with } u-r > 2.1$).

3.2 Distribution Shape Evolution

Seven equal-width redshift bins ($z = 0.002-0.35$, $\Delta z = 0.05$; $N = 3,900-34,000$). In each bin we compute skewness, excess kurtosis, and Box-Cox λ . Johnson SU parameters are reported in Appendix A; they show numerical instability and are excluded from the main analysis. We bootstrap 500 same-sized subsamples from the full pool to build 95% null envelopes and apply Spearman and Kendall rank tests for monotonic trend.

4. Results

4.1 Halo Mass Fully Mediates Colour–Environment for Ellipticals

Table 1 shows raw and partial Pearson correlations for the elliptical subsample. Raw $r = 0.07-0.09$; after controlling for $\log(\text{MassA})$ all partial correlations collapse to $r \approx 0$ ($p > 0.3$). At the group level ($N = 242$ groups) raw r rises to $0.13-0.15$ but again collapses after halo control.

Table 1. Partial correlation — ellipticals, controlling for $\log(\text{MassA})$

	Raw r	Raw p	Partial r	Partial p	Sig?
Individual level ($N = 1,270$)					
$\log(\text{CountInCyl})$	+0.066	0.019	−0.029	0.296	No
$\log(\text{DistanceTo5nn})$	−0.090	0.001	−0.006	0.824	No
$\log(\text{SurfaceDensity})$	+0.089	0.002	+0.005	0.869	No
$\log(\text{AGEDenPar})$	+0.071	0.012	+0.023	0.424	No
Group level ($N = 242$ groups)					
$\log(\text{CountInCyl})$	+0.133	0.039	+0.025	0.700	No
$\log(\text{DistanceTo5nn})$	−0.147	0.022	−0.046	0.480	No
$\log(\text{SurfaceDensity})$	+0.147	0.023	+0.044	0.492	No
$\log(\text{AGEDenPar})$	+0.037	0.572	−0.016	0.808	No

All partial correlations statistically indistinguishable from zero ($p > 0.3$).

4.2 Independent Environment Signal for the Full Population

For all galaxy types ($N = 22,531$ individuals; 5,775 groups), raw $r = 0.18-0.25$, remaining significant after halo control (partial $r = 0.10-0.16$, all $p < 10^{-55}$). Approximately 41–55% of the raw signal survives, indicating a secondary pathway.

Table 2. Partial correlation — full population, controlling for $\log(\text{MassA})$

	Raw r	Raw p	Partial r	Partial p	% retained
Individual level (N = 22,531)					
log(CountInCyl)	+0.240	8×10^{-2}	+0.109	7×10^{-1}	45.6%
log(DistanceTo5nn)	-0.245	3×10^{-3}	-0.124	4×10^{-3}	50.8%
log(SurfaceDensity)	+0.245	1×10^{-3}	+0.124	3×10^{-3}	50.8%
log(AGEDenPar)	+0.178	2×10^{-4}	+0.104	2×10^{-3}	58.4%
Group level (N = 5,775) — mean u-r					
log(CountInCyl)	+0.234	$< 10^{-3}$	+0.160	$< 10^{-3}$	68.4%
log(DistanceTo5nn)	-0.228	$< 10^{-3}$	-0.153	$< 10^{-3}$	67.1%
log(SurfaceDensity)	+0.228	$< 10^{-3}$	+0.153	$< 10^{-3}$	67.1%
log(AGEDenPar)	+0.154	$< 10^{-3}$	+0.128	$< 10^{-3}$	83.1%

All partial correlations remain highly significant ($p \ll 0.05$).

4.3 Colour Distribution Shape Evolves Monotonically with Redshift

Table 3 and Figure 1 show Box-Cox λ and skewness against redshift with bootstrap null envelopes. Both parameters lie outside the null at every bin. Skewness shifts from +0.90 at $z = 0.026$ to -0.82 at $z = 0.325$ (Spearman $r = -1.000$, $p = 4.0 \times 10^{-4}$). Box-Cox λ increases from -0.20 to +2.70 (Spearman $r = +1.000$, $p = 4.0 \times 10^{-4}$). The sign flip in skewness reflects the progressive consolidation of the red sequence. The transition through $\lambda = 0$ near $z \approx 0.08$ marks the approximate epoch at which the colour distribution crosses from sub-log-Normal to supra-Normal breadth.

Table 3. u-r distribution shape parameters by redshift bin

z centre	N	Box-Cox λ	Skewness	Excess kurtosis
0.026	3,932	-0.200	+0.899	+0.824
0.075	16,579	+0.431	+0.263	-0.750
0.125	29,589	+1.125	-0.148	-1.046
0.175	32,999	+1.694	-0.387	-0.897
0.225	27,116	+1.995	-0.523	-0.623
0.275	33,685	+2.532	-0.745	-0.188
0.325	21,588	+2.701	-0.821	+0.069

Spearman $r(\lambda, z) = +1.000$, $p = 4.0 \times 10^{-4}$; Spearman $r(\text{Skew}, z) = -1.000$, $p = 4.0 \times 10^{-4}$. All bins lie outside the bootstrap 95% null envelope.

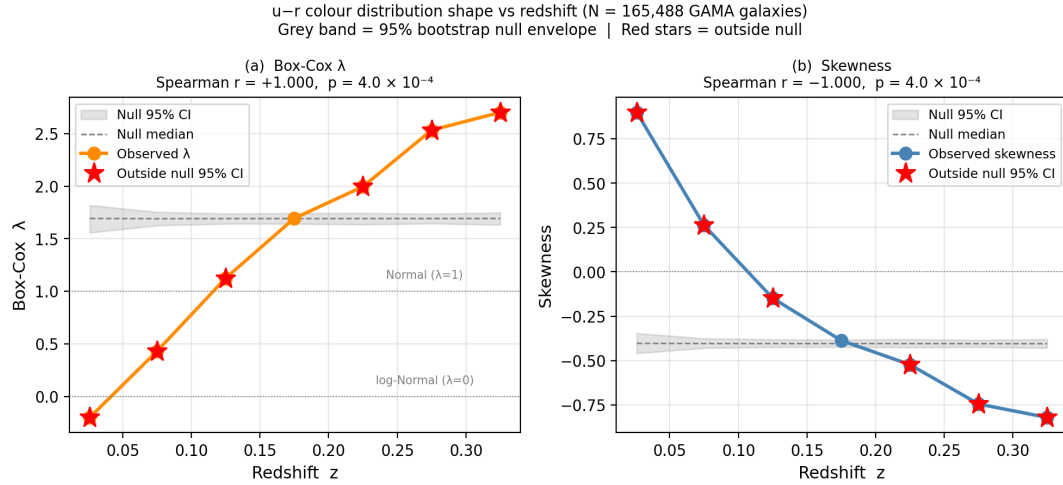


Figure 1. Box-Cox λ (left) and skewness (right) of the u-r colour distribution vs redshift. Grey band = 95% bootstrap null envelope (500 iterations). Red stars mark bins where the observed value lies outside the null. Both parameters show perfect monotonic trends (Spearman $|r| = 1.00$, $p = 4 \times 10^{-4}$).

5. Discussion

Our partial correlation results align with the picture that halo mass is the primary driver of colour–environment correlations for morphologically selected red galaxies (Peng et al. 2012; Woo et al. 2013). Complete mediation for ellipticals suggests that once a galaxy has settled into a massive halo and undergone morphological transformation, residual local environment adds little further quenching pressure. The ~50% survival of the signal in the full population points to secondary processes consistent with harassment and tidal stripping acting preferentially on late-type discs (Moore et al. 1996).

The distribution shape results offer a model-independent complement. Perfect monotonic Spearman trends ($|r| = 1.00$, $p < 0.001$) across seven independent redshift bins, combined with consistent departure from the bootstrap null, constitute strong evidence of real physical evolution — a direct quantitative answer to the objection that shape differences merely reflect statistical variation.

6. Conclusions

1. Halo mass fully mediates colour–environment for elliptical galaxies (N = 1,270); partial correlations collapse to $r \approx 0$ after controlling for $\log(\text{MassA})$.
2. For the full galaxy population (N = 22,531), ~50% of the signal survives halo control, indicating a morphology-linked secondary pathway.
3. Skewness and Box-Cox λ evolve monotonically with redshift ($|\text{Spearman } r| = 1.00$, $p = 4 \times 10^{-4}$): skewness from +0.90 to −0.82 and λ from −0.20 to +2.70, inconsistent with sampling noise and consistent with progressive consolidation of the red sequence.

Data Availability

GAMA DR4: <https://www.gama-survey.org/dr4/>. Analysis notebooks available on request.

Appendix A: Johnson SU Fit Parameters

Table A1 gives the full set of shape parameters including Johnson SU (*a*, *b*). The JSU *a* parameter shows large non-monotonic variation despite the smooth evolution of skewness and λ , arising from correlation between *a* and *b* on a near-flat likelihood ridge.

Table A1. Full shape parameters including Johnson SU

z	N	JSU a	JSU b	Box-Cox λ	Skewness	Kurt
0.026	3,932	−2.763	+2.210	−0.200	+0.899	+0.824
0.075	16,579	−11.677	+5.925	+0.431	+0.263	−0.750
0.125	29,589	+20.679	+7.171	+1.125	−0.148	−1.046
0.175	32,999	+9.776	+3.566	+1.694	−0.387	−0.897
0.225	27,116	+3.431	+2.112	+1.995	−0.523	−0.623
0.275	33,685	+2.802	+1.752	+2.532	−0.745	−0.188
0.325	21,588	+2.836	+1.802	+2.701	−0.821	+0.069

JSU *a*: Spearman $r = +0.29$, $p = 0.53$ (not significant). Box-Cox λ and skewness: $|r| = 1.00$, $p < 0.001$. JSU *a* varies by >30× while skewness changes by only 0.75 — instability is numerical, not physical.

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